Lab 5: Data Visualization with ggplot

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Background

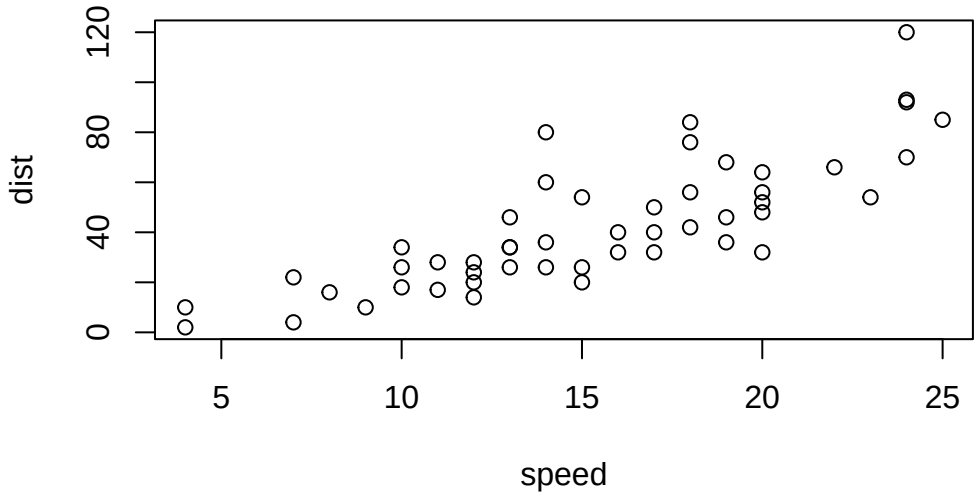
R has multiple plotting systems. This is a brief comparison of base R graphics and **ggplot2** using the built-in `cars` dataset.

```
head(cars)
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

Base R

```
plot(cars)
```



ggplot2

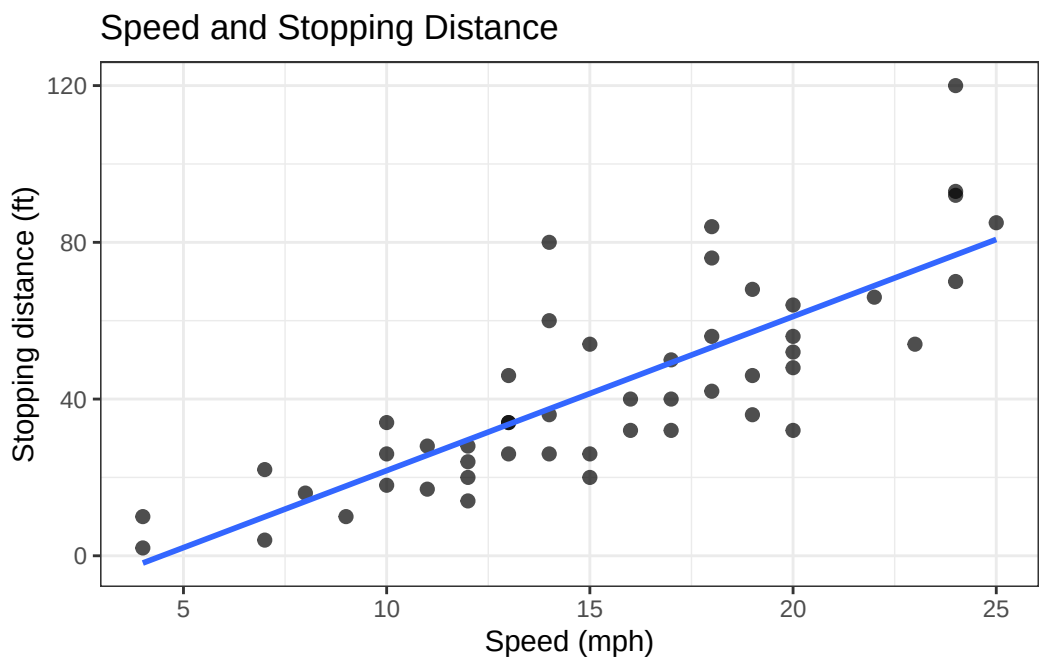
Install once in the Console: install.packages("ggplot2") Load each session with library():

```
library(ggplot2)
ggplot(cars)
```

A ggplot typically includes: 1. data 2. aesthetics (aes()) 3. one or more geoms

```
ggplot(cars) +
  aes(x = speed, y = dist) +
  geom_point(alpha = 0.7, size = 2) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    title = "Speed and Stopping Distance",
    x = "Speed (mph)",
    y = "Stopping distance (ft)"
  ) +
  theme_bw()
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Gene Expression Plot

Load gene expression data comparing control and drug-treated samples:

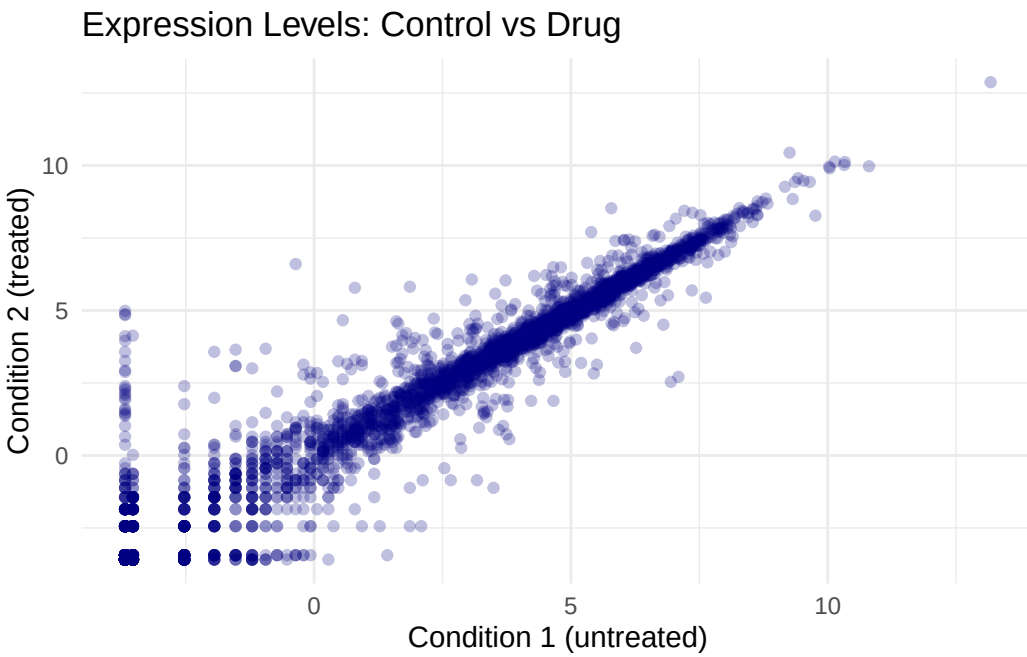
```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

```
        Gene Condition1 Condition2      State
1      A4GNT -3.6808610 -3.4401355 unchanging
2       AAAS  4.5479580  4.3864126 unchanging
3      AASDH  3.7190695  3.4787276 unchanging
4       AATF  5.0784720  5.0151916 unchanging
5       AATK  0.4711421  0.5598642 unchanging
6 AB015752.4 -3.6808610 -3.5921390 unchanging
```

Simple scatter plot

```
ggplot(genes) +
  aes(x = Condition1, y = Condition2) +
  geom_point(color = "navy", alpha = 0.25, size = 1.5) +
```

```
  labs(
    title = "Expression Levels: Control vs Drug",
    x = "Condition 1 (untreated)",
    y = "Condition 2 (treated)"
  ) +
  theme_minimal()
```



Check counts for each expression state:

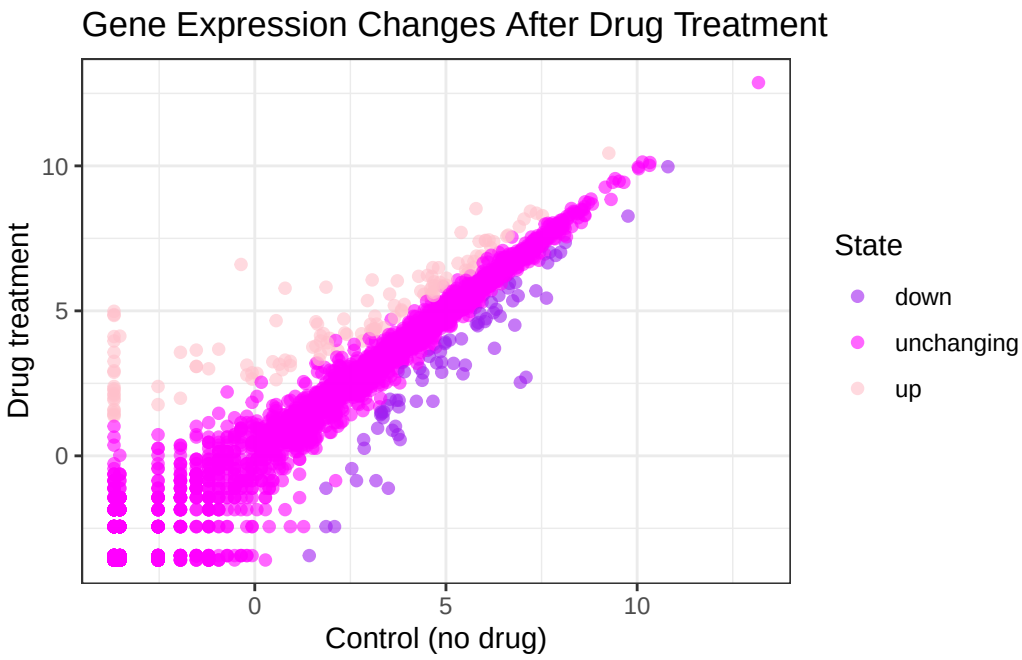
```
table(genes$State)
```

```
      down unchanging         up
        72       4997        127
```

Color points by expression state:

```
ggplot(genes) +
  aes(x = Condition1, y = Condition2, col = State) +
  geom_point(alpha = 0.6, size = 1.7) +
  scale_color_manual(values = c("purple", "magenta", "pink")) +
```

```
  labs(
    title = "Gene Expression Changes After Drug Treatment",
    x = "Control (no drug)",
    y = "Drug treatment"
  ) +
  theme_bw()
```



Gapminder exploration

Exploring the gapminder dataset:

```
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv
gapminder <- read.delim(url)
head(gapminder)
```

```
      country continent year lifeExp      pop gdpPercap
1 Afghanistan      Asia 1952  28.801  8425333  779.4453
2 Afghanistan      Asia 1957  30.332  9240934  820.8530
3 Afghanistan      Asia 1962  31.997 10267083  853.1007
4 Afghanistan      Asia 1967  34.020 11537966  836.1971
5 Afghanistan      Asia 1972  36.088 13079460  739.9811
6 Afghanistan      Asia 1977  38.438 14880372  786.1134
```

How many rows are in the dataset?

```r
nrow(gapminder)
```

```
[1] 1704
```

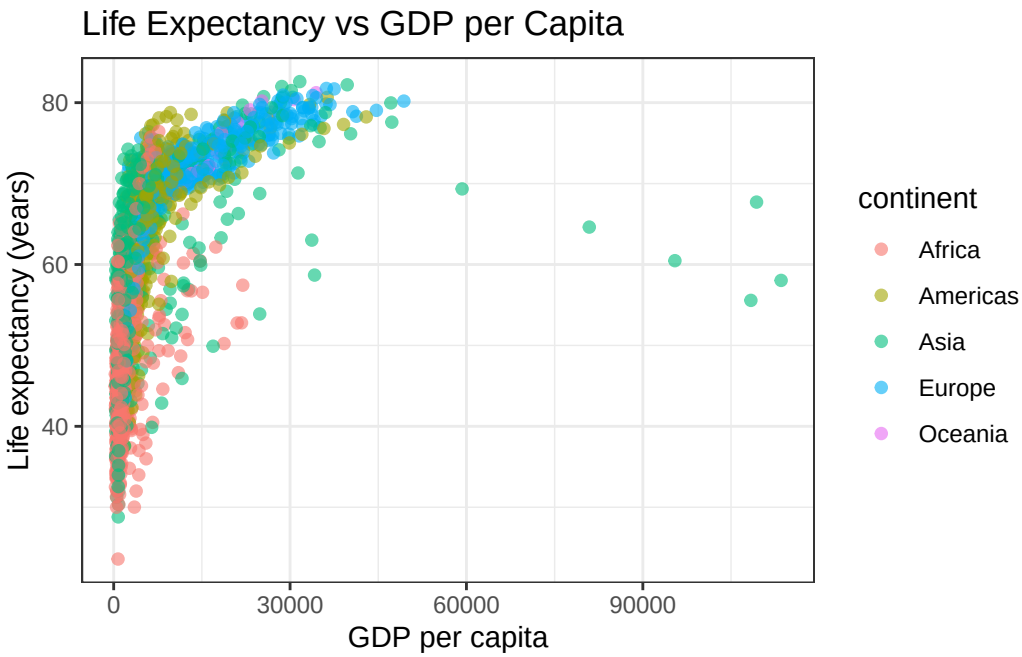
How many continents are represented?

```r
table(gapminder$continent)
```

```
  Africa Americas     Asia   Europe  Oceania 
     624      300      396      360       24 
```

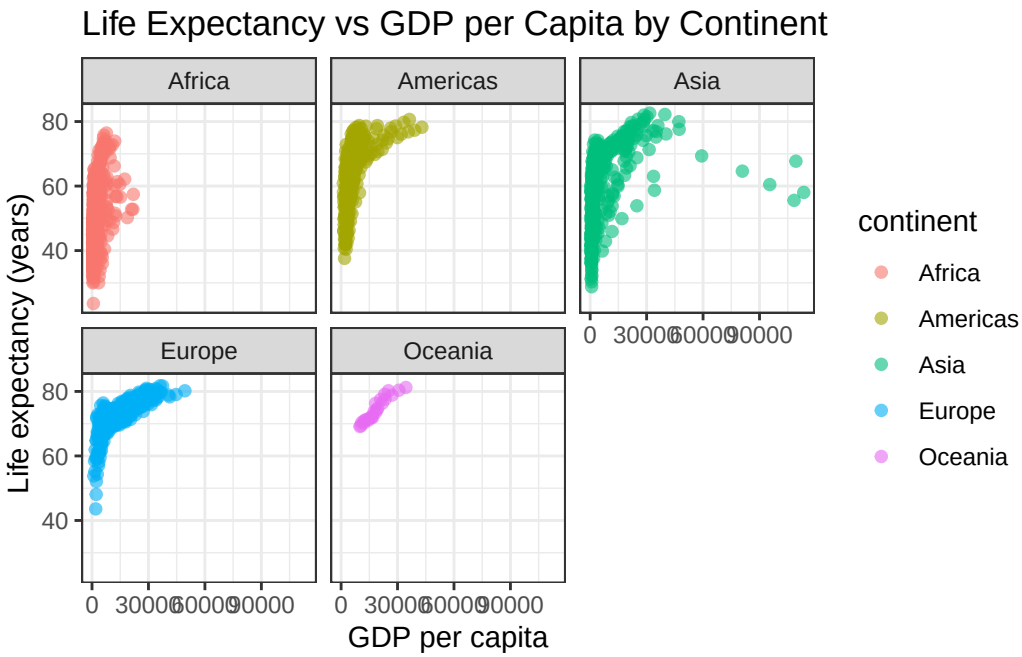
GDP per capita vs life expectancy

```r
ggplot(gapminder) +
  aes(x = gdpPercap, y = lifeExp, col = continent) +
  geom_point(alpha = 0.6, size = 1.7) +
  labs(
    title = "Life Expectancy vs GDP per Capita",
    x = "GDP per capita",
    y = "Life expectancy (years)"
  ) +
  theme_bw()
```



Facet by continent:

```
ggplot(gapminder) +
  aes(x = gdpPercap, y = lifeExp, col = continent) +
  geom_point(alpha = 0.6, size = 1.7) +
  facet_wrap(~ continent) +
  labs(
    title = "Life Expectancy vs GDP per Capita by Continent",
    x = "GDP per capita",
    y = "Life expectancy (years)"
  ) +
  theme_bw()
```



First look at dplyr

The dplyr package provides functions for data manipulation, including filter().

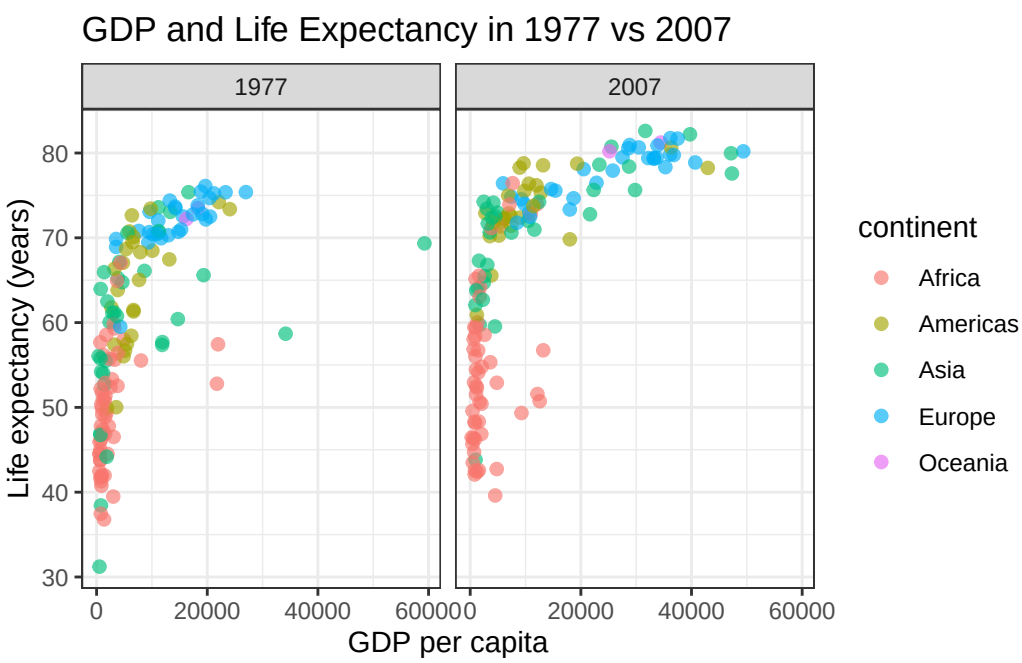
```
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

Example: Ireland in 2007

```r
filter(gapminder, year == 2007, country == "Ireland")
```

```
  country continent year lifeExp     pop gdpPercap
1 Ireland    Europe 2007  78.885 4109086     40676
```

Compare two years using faceting:

```r
input <- filter(gapminder, year == 2007 | year == 1977)

ggplot(input) +
  aes(x = gdpPercap, y = lifeExp, col = continent) +
  geom_point(alpha = 0.65, size = 1.9) +
  facet_wrap(~ year) +
  labs(
    title = "GDP and Life Expectancy in 1977 vs 2007",
    x = "GDP per capita",
    y = "Life expectancy (years)"
  ) +
  theme_bw()
```